

HEALTHCONNECT CLINIC — WEEK 5 ANALYTICS REPORT

1. Data preparation — what I checked before analysing

The file has 5,000 appointments and 18 columns. Each row is one appointment, and the same patient can appear more than once, which I will use later for the history analysis.

Before making any chart, I ran four checks:

Missing values. `distance_to_clinic_km` is blank in 90 rows and `waiting_time_minutes` in 60 rows (for example HC-00039 has no distance). I decided not to fill these with averages — guessing a patient's distance could create a fake pattern — so I kept the blanks and simply left those rows out of the distance and waiting-time charts.

Duplicates. None. `appointment_id` runs from HC-00001 to HC-05000 with no repeats, so every row really is one appointment.

Consistency. Whenever `reminder_sent` = No, the channel shows "None". That means "None" is not missing data — it just means no reminder was sent, so I treated it that way. I also checked that `previous_no_shows` never exceeds `previous_appointments`, and that every age sits inside its `age_group` band. Both checks passed, which made me trust the file.

Dates. They arrived as text (e.g. 2/6/2025), so I converted them to real dates before doing anything time-based.

2. What the data says

First, the outcome split:

No-Show: 2,423 (48.5%)

Attended: 2,314 (46.3%)

Cancelled: 263 (5.3%)

Almost half of all booked appointments end in a no-show. That told me the clinic's real problem is people not turning up — not cancellations.

Then I tested one factor at a time:

Booking lead time is the strongest pattern I found. The no-show rate climbs steadily: 27.8% when booked within a week, 33.6% at 8–14 days, 43.2% at 15–30 days, and 60.5% at 31–60 days. On average, missed appointments were booked 34.5 days ahead, against 24.5 days for attended ones.

Past behaviour matters. Patients who have missed before miss at 55.4%, against 43.5% for patients with a clean history.

Distance matters too. The no-show rate rises from 46.5% (0–10 km) to 68.1% (30+ km).

Reminders help, but only a little. 47.4% no-show with a reminder vs 51.4% without. Among channels, SMS does best (45.8%) and WhatsApp worst (49.8%).

Type, day and time of day made almost no difference (all between 46% and 51%), so I will not build recommendations around them.

3. The KPIs I will track

Kpi	How I calculate	What it tells clinic	My results
No show rate	no-shows ÷ all appointment	The size of the problem	48.5%
Reminder Effectiveness	no-show rate with vs without reminder	Whether reminders are worth the effort	47.4% vs 51.4%
Repeat No-Show Rate	no-show rate among patients with 1+ past no-show	Whether a small group causes most of the problem	55.4% (vs 43.5% clean)
Avg. Booking Lead Time	average lead days per outcome	How early booking relates to missing	34.5 vs 24.5 days
Distance Band	no-show rate per distance group	Whether travel distance is a barrier	46.5% → 68.1%

4. What I would tell the clinic

Re-confirm long-lead bookings. Appointments booked 31–60 days ahead miss at 60.5%. A confirmation message at the two-week mark and again 72 hours before would target the riskiest group.

Give extra attention to repeat missers. Patients with past no-shows miss at 55.4%. A phone call, not just an SMS, is worth trying for them.

Help far-away patients. At 30+ km the no-show rate is 68.1%. Offering telehealth for follow-ups, or rescheduling to closer slots, could recover many of these.

Do not over-invest in reminders alone. They only move the rate by about 4 points. Necessary, but not enough on their own.

Do not redesign the schedule around day, time or type. The differences are too small to act on.

5. Limits

The 90 missing distance values mean the distance figures exclude some patients.

I cannot see reasons like illness or transport problems, so some no-shows stay unexplained.

Correlation is not causation — the clinic should pilot these changes before rolling them out fully.